Immigration, Crime, and Crime (Mis)Perceptions: README and Guidance

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Overview

The code in this replication package generates the data and results from the three main data sources: official immigration data, official homicide data, and a rich annual victimization survey. The software used is Stata. The master do File run all of the codes to generate the data for the figures and tables in the paper. The replicator should expect the code to run for about 1 hour.

Data Availability and Provenance Statements

This section provides information on each data source.

1. Immigration Data

Data type: Public Use data

We obtained individual-level data on all visa and permanent residence permits granted by the Chilean Department of State. This data includes basic demographic statistics such as date of birth, nationality, municipality of residence at time of application, gender, and self-reported variables on education and labor market experience. Importantly, our data on immigration comes from administrative records and includes authorized immigration only. The appropriate citation for this data is:

- Department-of-State. 2005–2018. "**Immigration Data from Chilean Department of State 2005-2018**" Department of State, Government of Chile, https://www.extranjeria.gob.cl/estadisticas-migratorias/ (accessed December 10, 2019).

Since these are data on immigrant flows and we need data on immigrant stocks, we use the 2002 Chile Census from (National Institute of Statistics, INE) to construct the initial stocks of immigrants by municipality. The appropriate citation for this data is:

- National Institute of Statistics. 2002. "**2002 Chile Census.**" National Institute of Statistics (INE), Government of Chile. Access through transparency law by formal request at Consejo para la Transparencia platform, https://www.portaltransparencia.cl/PortalPdT/ingreso-sai-v2?idOrg=undefined (accessed December 10, 2019).

2. Official Homicide Data

Data type: Confidential data

We obtained data from the Chilean Ministry of Internal Affairs (*Ministerio del Interior*) on all homicides perpetrated during 2005-2017. The homicide rate in Chile was approximately 3.46 per 100,000 inhabitants in 2008 and increased slightly to 3.57 by the end of 2017. Upon special request, this dataset includes the nationality of the alleged perpetrator. This is an important piece of information since we examine whether the immigrant shock affects the total homicide rate and whether the composition of perpetrators changed.

The homicides data used in Tables A1 (last row), V, X, A4 (Panel A.4 and Panel B.4), A5 (Column 4), XI (Column 4), XII (Column 4), XIII (Column 4), A16 (Column 4), A.2, A.3, A.8 (Panel B), A.9 (Column 2), A.12 (Column 8), and A.15 (Column 8), as well as in Figures VI and VII (Panel b) cannot be available in an openly accessible data repository like the one required by AEJ: Applied. The homicides data was obtained under a Non-Disclosure Agreement (NDA) with Ministerio del Interior (Ministry of Internal Affairs), which is the way researchers must proceed to get access to it. We can help researchers to get access to the original files. The appropriate citation for this data is:

- Ministry-of-Internal-Affairs. 2005-2017. "**Official Homicide Data from Chilean Police 2005-2017**". Confidential data obtained through a Non-Disclosure Agreement with Ministry of Internal Affairs, Government of Chile, (accessed August 15, 2021).

3. Victimization Data

Data type: Public Use data

We harmonized a set of variables dealing with crime perceptions, behavior related to the adoption of security measures, and victimization that were included in the ENUSC (*Encuesta Nacional Urbana de Seguridad Ciudadana*) survey. This is an annual household survey covering the period 2008-2017, for which the field work took place between October and December of each year. There do exist previous editions of the ENUSC for the years 2003, 2005, and 2006, but the methodology underwent a series of changes in 2008, rendering comparisons of the pre- and post-2008 data uninformative. The survey was also conducted in 2018 but, unlike previous years, the municipality codes were not made available to researchers. Although the ENUSC only covers a subset of municipalities (101 out of 346 municipalities in Chile), these municipalities represent approximately 80% of the national population; a proportion that is even larger (around 95 percent) for the immigrant population, who are more likely to live in large urban areas. Overall, the ENUSC analysis data comprises almost 250,000 observations distributed across 101 municipalities throughout the 2008-2017 period. The appropriate citation for this data is:

- National Institute of Statistics 2003-2017 "**Encuesta Nacional Urbana de Seguridad Ciudadana (ENUSC), 2003-2017**". National Institute of Statistics (INE), Government of Chile, https://www.ine.cl/estadisticas/sociales/seguridad-publica-y-justicia/seguridad-ciudadana (accessed December 10, 2019)

4. Regional Population Data

Data type: Public Use data

Regional population information comes from INE (*National Institute of Statistics*) estimates using projections from census data, which allows us to calculate the rate of immigration by municipality for each year. The appropriate citation for this data is:

- National Institute of Statistics. 2002-2035. "**Regional Population Data 2002-2035.**" National Institute of Statistics (INE), Government of Chile, https://www.ine.cl/estadisticas/sociales/demografia-y-vitales/proyecciones-de-poblacion (accessed December 10, 2019)

5. Bilateral Flows of International Migrants Data

Data type: Public Use data

We use UN Population Division migration data, which include data on bilateral flows of international migrants for 45 countries, to construct the Bartik Instrument. Particularly, we use it to construct the log change of immigrants of different nationalities in destination countries other than Chile, where the variation in this term is by construction orthogonal to demand-pull factors embedded in Chilean municipalities. For most nations, information is available for at least 1990 to 2017 at a five-year frequency (e.g., 1990, 1995, 2000, 2005, 2010, 2015, as well as 2017), and includes both inflows and outflows according to place of birth, citizenship, and place of previous/next residence for both foreigners and nationals. Though coverage is limited to the most relevant origin-destination cells, we were able to build the 2010-2017 (log) changes for 11 countries: Argentina, Bolivia, Brazil, China, Colombia, Ecuador, Haiti, Peru, Spain, USA, and Venezuela17. Collectively, these cells represent 87% and 94% of residence permits in 2008 and 2017, respectively. The appropriate citation for this data is:

- United Nations. 1990-2017. **"Bilateral Flows of International Migrants Data, 1990-2017."** United Nations Population Division, https://www.un.org/en/development/desa/population/migration/data/estimates2/data/UN_MigrantStockTotal_2017.xlsx (accessed December 10, 2019)

6. Bilateral Ethnic/Genetic Distance Data

Data type: Public Use data

A plausible hypothesis is thus the farther the ethnic distance between natives and immigrants, the greater the prejudice and fear. To measure ethnic bilateral relatedness, we employ the genetic distance between two countries. We use a measure of bilateral ethnic/genetic distance to compute two outgroup indicators: the average ethnic distance between Chileans and immigrants arriving in different municipalities at different times and the average ethnic distance of immigrants relative to Europeans. This measure is constructed by Spolaore and Wacziarg (2018). Their work is, in turn, based on the bilateral genetic distances between populations calculated by Cavalli-Sforza et al. (1994) and then extended by Pemberton, DeGiorgio and Rosenberg (2013). The appropriate citations for this data are:

- Cavalli-Sforza, Luigi Luca, Luca Cavalli-Sforza, Paolo Menozzi, and Alberto Piazza. 1994. The History and Geography of Human Genes. Princeton University Press

- Pemberton, Trevor J, Michael DeGiorgio, and Noah A Rosenberg. 2013. "Population structure in a comprehensive genomic data set on human microsatellite variation." G3: Genes, Genomes, Genetics, 3(5): 891-907.

- Spolaore, Enrico, and Romain Wacziarg. 2018a. "Ancestry and development: New evidence." Journal of Applied Econometrics, 33(5): 748-762.

- Spolaore, Enrico, and Romain Wacziarg. 2018b. "Data for: Ancestry and development: New evidence." Journal of Applied Econometrics, 33(5): 748–762. http://qed.econ.queensu.ca/jae/datasets/spolaore001/.

7. Media Data

Data type: Public Use data

In order to better understand the public misperception of the effects of immigration on crime, we also explore the potential role of local media. The Chilean Department of Telecommunications (SUBTEL) provides a detailed dataset on local radio broadcasting at the municipality level, which we use to create a measure of exposure. We use it to divide municipalities into two groups: ``low media," where the number of local radio stations per capita is below the median, and ``high media" otherwise. The appropriate citation for this data is:

- SUBTEL. 2019. "**SUBTEL Media Data."** Undersecretary of Telecomunications, Government of Chile, https://www.subtel.gob.cl/inicio-concesionario/servicios-de-telecomunicaciones/servicios-de-radiodifusion-sonora/ (accessed December 10, 2019)

8. Additional Controls

Data type: Public Use data

For robustness purposes, in some of our regressions we include controls at the municipality level taken from the Chile National Socioeconomic Characterization Survey (CASEN). This household survey covers the entire country and includes standard questions related to demographics, labor market outcomes, income, and education. During the analysis period (2008-2017), it was conducted in 2009, 2011, 2013, 2015, and 2017. The appropriate citation for this data is:

- Ministry-of-Social-Development. 2000-2017- "Encuesta CASEN 2000-2017." Ministry of Social Development, Government of Chile, http://observatorio.ministeriodesarrollosocial.gob.cl/encuesta-casen (accessed December 10, 2019)

9. Unique Territorial Codes (CUT) Data

Data type: Public Use data

CUT data is used to link different data sources at the Municipality level. Some datasets in DTA containing the Unique Territorial Codes of the communes and the regions have been generated manually. The appropriate citation for this data is:

- **SUBDERE. 2019. "Unique Territorial Codes (CUT) Data."** Undersecretary of Regional Development (SUBDERE), Government of Chile, https://www.subdere.gov.cl/documentacion/códigos-únicos-territoriales-actualizados-al-06-de-septiembre-2018 (accessed December 10, 2019)

Statement about Rights

I certify that the author(s) of the manuscript have legitimate access to and permission to use the data used in this manuscript.

I certify that the author(s) of the manuscript have documented permission to redistribute/publish the data contained within this replication package, except for the Official Homicide Data which requires a special request to the *Chilean Ministry of Internal Affairs* be filed by researchers.

Summary of Availability

Almost all the datasets used in this analysis can be publicly available since there are outputs obtained from combining public use datasets. However, those datasets that are outputs obtained from using homicide dataset cannot be publicly available.

- Homicide data and datasets that are outputs obtained from using it **cannot be made** publicly available.

Details on each Data Source

This section provides information on how each data source is obtained and where we provide it in the repository.

Confidential data source:

Homicide Data from Chilean Police (2005-2017):
Homicide data is confidential and cannot be shared or shared in the public domain.
We requested the data to Ministerio del Interior (Ministry of Internal Affairs). Under the transparency law, researchers can file a special request at https://www.interior.gob.cl/transparenciaactiva/.

Public use data used by the authors:

Immigration Data from Chilean Department of State (2005-2018)

Obtained from Chilean Department of State (*Extranjería*) website:
https://www.extranjeria.gob.cl/estadisticas-migratorias/

The raw data sets used are provided in the public use data folder
(**\Replication\data\raw\immigration_Correciones.dta**)

2002 Chile Census 2002

Accesing this data requires to fill a formal request to National institute of Statistics (INE) through the Consejo para la Transparencia platform in this website:
https://www.portaltransparencia.cl/PortalPdT/ingreso-sai-v2?idOrg=undefined.

The raw data set used is provided in the public use data folder.
(**\Replication\data\raw\PERSONA_comuna_distrito.dta**)

ENUSC (2003-2017)

Most microdata surveys can be downloaded from INE website at:
https://www.ine.cl/estadisticas/sociales/seguridad-publica-y-justicia/seguridad-ciudadana

If any of the cross-sectional surveys is not available to download from the INE website, researchers can file an information request at:
https://www.portaltransparencia.cl/PortalPdT/ingreso-sai-v2?idOrg=1003

Datasets that researchers need to ask are the following:
Base-de-datos-xiv-enusc-2017.rar
9. ENUSC XIII Base Usuario 2016.sav.zip
12. ENUSC XII Base Usuario 2015.sav.zip
11. ENUSC XI Base Usuario 2014.sav.zip
10. ENUSC X Base Usuario 2013.sav.zip
09. ENUSC IX Base Usuario 2012.sav.zip
08. ENUSC VIII Base Usuario 2011.sav.zip
07. ENUSC VII Base Usuario 2010.sav.zip
06. ENUSC VI Base Usuario 2009.sav.zip
05. ENUSC V Base Usuario 2008.sav.zip
04. ENUSC IV Base Usuario 2007.sav.zip
03. ENUSC III Base Usuario 2006.sav.zip
02. ENUSC II Base Usuario 2005.sav.zip
01. ENUSC I Base Usuario 2003.sav.zip

The raw data sets used are provided in the public use data folder.
(**Replication\data\raw\ENUSC)**

Regional Population Data (2002, 2035)

These data are publicly available from the following link:
https://www.ine.cl/estadisticas/sociales/demografia-y-vitales/proyecciones-de-poblacion.

The raw data set used is provided in the public use data folder.
(**Replication\data\raw\estimaciones-y-proyecciones-2002-2035-comunas.xlsx)**. In order to link different data sources, some datasets in DTA containing the populations of the communes in 2010 have been generated manually.

Bilateral Flows of International Migrants Data (1990, 1995, 2000, 2005, 2010, 2015, 2017)

Data is publicly available at this link:
https://www.un.org/en/development/desa/population/migration/data/estimates2/data/UN_MigrantStockTotal_2017.xlsx

The raw data set used is provided in the public use data folder.
(**Replication\data\raw\UN_MigrantStock.xlsx**). In that file (that we then use in Stata) we manually reshaped the original raw data so to have a double entry table of countries/countries, one table per year (1990, ..., 2017).

Bilateral Ethnic/Genetic Distance Data (2018)

Data is publicly available at this link:
http://qed.econ.queensu.ca/jae/datasets/spolaore001/

The raw data set used is provided in the public use data folder. (**Replication\data\raw\bilateral_replic.dta**)

SUBTEL Media Data (2019)

Data is publicly available at this link:
https://www.subtel.gob.cl/inicio-concesionario/servicios-de-telecomunicaciones/servicios-de-radiodifusion-sonora/

The file should be downloaded clicking on “Descargar archivo en formato XLS”.

Caveat: data is updated every year and therefore there could be small differences. Our dataset was downloaded in December 10, 2019. To obtain the exact version we use in the paper, researchers should email SUBTEL asking for the same file ("Servicios de Radiodifusion Sonora: Información Relativa a Concesiones") by December 10, 2019. The data is public.

The raw format is XLS. We manually categorized each radio station according to the column "Comuna Planta", which contains the name of the municipality. Then we counted the number of broadcasters by municipality, which is what appears in the media file we use.

The raw data set used is provided in the public use data folder.
(\Replication\data\raw\countTS.dta)

CASEN (2000, 2003, 2006, 2009, 2011, 2013, 2015, 2017)

The CASEN data sets can be downloaded from the following website
http://observatorio.ministeriodesarrollosocial.gob.cl/encuesta-casen.

The raw data sets used are provided in the public use data folder.
(**\Replication\data\raw\casen*año*.dta, *año*=2000, 2003, 2006, 2009, 2011, 2013, 2015, 2017**)

Unique Territorial Codes (CUT) Data
The raw data set used is provided in the public use data folder. (**\Replication\data\raw\División-Político-Administrativa-y-Servicios-de-Salud-Histórico.xlsx**). In order to link different data sources, some datasets in DTA containing the Unique Territorial Codes of the communes and the regions have been generated manually.

An updated version of CUT data can be downloaded from:
https://www.subdere.gov.cl/documentacion/códigos-únicos-territoriales-actualizados-al-06-de-septiembre-2018

Computational requirements

Software Requirements

All analyses in the paper are run using Stata MP 16. The following packages are needed to run the code:

- groups
- outreg2
- ivreg2
- reg_ss

- ivreg_ss
- weakiv
- ranktest
- avar
- sg97_5

Memory and Runtime Requirements

Summary

Approximate time needed to reproduce the analyses on a standard desktop machine: 1-2 hours.

Details

The code was last run on a on a Windows 10 Home 64-bit Operating System, X64 based server with 12 GB of RAM. Computation took about 1 hour.

Instructions to Replicators

To replicate, run the Master do File, changing the path in line 5. Running the Master do File will run all the commands to generate the data to replicate all the tables and figures in the paper. The last row of tables V and X (Mean DV) needs to be manually added (line 39 of "Tables_V_A2.do" displays the results in these rows).

Description of programs/code

- Program in `code/0_stata_setup` will add required packages from SSC.
- Programs in `code/1_create` will extract and reformat all datasets referenced above.
- Programs in `code/2_analysis will` generate all tables and figures in the paper. Each program identifies the table or figure it creates (e.g., `Table_XII.do`). Output files are called appropriate names (`results/tables/Table_XII.tex`) and should be easy to correlate with the manuscript.

To run the codes `code\2_analysis\Figures_A2_A3_A4.do` and `code/2_analysis/Figure_A5.do`, it is necessary to install the program 'bartik_weight' provided by Goldsmith-Pinkham, Sorkin and Swift (2020). (Goldsmith-Pinkham, Paul, Isaac Sorkin, and Henry Swift. 2020. "Bartik instruments: What, when, why, and how." American Economic Review, 110(8): 2586-2624). Master do file automatically installs this package. To install this Stata package manually, it is necessary to clone or download the repo available at this link https://github.com/paulgp/bartik-weight, and then copy bartik_weight.ado and bartik_weight.sthlp to your personal ado folder. You can find this folder using the *sysdir* command. Both bartik_weight.ado and bartik_weight.sthlp are provided as part of the replication file ("…\Replication\ado").

List of final datasets and programs

The code generates the final data needed to replicate the tables and figures in the paper.

ENUSC.do

This do-file generates the ENUSC consolidated base (2008-2017).

Data_Table_I.do

This do-file generates the final base "TableI.dta". This base is used to replicate Table A1.

OLS_1.do

This do-file generates the final base "enusc_OLS.dta". This base is used to replicate Tables A1, I, II, III, and IV, and Figure A6 (Panels a and c).

IV_1.do

This do-file generates the final base "enusc_IV.dta". This base is used to replicate Tables VI, VII, VIII, IX, A4 (Panel A.1-A.3), XIII (Columns 1-3), A.6, A.7, A.8 (Panel A) and A.9 (Columns 1 and 3).

Pretrends.do

This do-file generates the final base "pretrendsfinalbase.dta". This base is used to replicate Figures A2, A3 y A4.

IV_2.do

This do-file generates the final base "IV_panel.dta". This base is used to replicate Table A4 (Panel B.1-B.3).

OLS_2.do

This do-file generates the final base "channels_OLS.dta". This base is used to replicate Tables XI (Columns 1-3), XII (Panel A, Columns 1-3) and XVII (Columns 1-3).

OLS_3.do

This do-file generates the final base "channels_OLS2.dta". This base is used to replicate Table XII (Panel B, Columns 1-3).

OLS_4.do

This do-file generates the final base "channels_OLS3.dta". This base is used to replicate Table A16 (Columns 1-3).

IV_3.do

This do-file generates the final base "enusc_IV_haiperven.dta". This base is used to replicate Tables A.10, A.11 and A.12 (Columns 1-7).

IV_4.do

This do-file generates the final base "enusc_IV_share2002.dta". This base is used to replicate Tables A.13, A.14 and A.15 (Columns 1-7).

The following programs require confidential data.

OLS_1_homicides.do

This do-file generates the final base "homicides_OLS.dta". This base is used to replicate Tables A1 (last row), V and A.2, and Figure A6 (Panel b).

IV_1_homicides.do

This do-file generates the final base "homicides_IV.dta". This base is used to replicate Tables X, XI (Panel A.4), XII (Column 4), A.3, A.8 (Panel B) and A.9 (Column 2).

Pretrends_homicides.do

This do-file generates the final base "pretrendsfinalbase_homicides.dta". This base is used to replicate Figure A5.

IV_2_homicides.do

This do-file generates the final base "IV_panel_homicides.dta". This base is used to replicate Table A4 (Panel B.4).

OLS_2_homicides.do

This do-file generates the final base "channels_OLS_homicides.dta". This base is used to replicate Tables XI (Column 4), XII (Panel A, Column 4) and XIII (Column 4).

OLS_3_homicides.do

This do-file generates the final base "channels_OLS2_homicides.dta". This base is used to replicate Table XII (Panel B, Column 4).

OLS_4_homicides.do

This do-file generates the final base "channels_OLS3_homicides.dta". This base is used to replicate Table A16 (Column 4).

IV_3_homicides.do

This do-file generates the final base "homicides_IV_haiperven.dta". This base is used to replicate Table A.12 (Column 8).

IV_4_homicides.do

This do-file generates the final base "homicides_IV_share2002.dta". This base is used to replicate Table A.15 (Column 8)

List of tables and programs

The provided code reproduces all tables and figures in the paper.

Figures_I_A1.do

This do-file creates Figures I and A1.

Table_A1.do

This do-file creates Table A.1, but without homicide data.

Table_A1_homicides.do

This do-file creates Table A.1 for homicide data. It requires confidential data.

Tables_I_II_III_IV.do

This do-file creates Tables I, II, III and IV.

Tables_V_A2.do

This do-file creates Tables V and A.2. It requires confidential data.

Tables_VI_VII_VIII_IX.do

This do-file creates Tables VI, VII, VIII and IX.

Tables_X_A3.do

This do-file creates Tables X and A.3. It requires confidential data.

Figures_A2_A3_A4.do

This do-file creates graphs for Figures A2, A3 and A4.

Figure_A5.do

This do-file creates graphs for Figure A5. It requires confidential data.

Table_A4.do

This do-file creates Table A4 (Panels A.1-A.3 and B.1-B.3).

Table_A4_homicides.do

This do-file creates Table A4 (Panels A.4 and B.4). It requires confidential data.

Table_A5.do

This do-file creates Table A5 (Columns 1-3).

Table_A5_homicides.do

This do-file creates Table A5 (Column 4). It requires confidential data.

Tables_XIV_XV_XVI_XVII.do

This do-file creates Tables XIV (Columns 1-3), XV (Panels A and B, Columns 1-3), XVI (Columns 1-3) and XVII (Columns 1-3).

Tables_XI_XII_XIII_A16_homicides.do

This do-file creates Tables XI (Column 4), XII (Panels A and B, Column 4), XIII (Column 4) and A16 (Columns 4). It requires confidential data.

Figure_A6_a_c.do

This do-file creates Figure A6 (panels (a) and (c)).

Figure_A6_b.do

This do-file creates Figure A6 (panel (b)). It requires confidential data.

Tables_A6_A7_A8PanelA.do

This do-file creates Tables A.6, A.7 and A.7 (Panel A).

Table_A8PanelB.do

This do-file creates Table A.8 (Panel B). It requires confidential data.

Table_A9_Cols1and3.do

This do-file creates Table A.9 (Columns 1 and 3).

Table_A9_Col2.do

This do-file creates Table A.9 (Column 2). It requires confidential data.

Tables_A10_A11_A12.do

This do-file creates Tables A.10, A.11 and A.12 (Columns 1-8).

Table_A12_homicides.do

This do-file creates Table A.12 (Column 9). It requires confidential data.

Tables_A13_A14_A15.do

This do-file creates Tables A.13, A.14 and A.15 (Columns 1-8).

Table_A13_homicides.do

This do-file creates Table A.15 (Column 9). It requires confidential data.